

MySQL - db_43xxeqqag



```

18 START TRANSACTION;
19 INSERT INTO StudentEnrollments (enrollment_id, student_name, course_id, enrollment_date)
20 VALUES (4, 'Ashish', 'CSE101', '2024-07-02');
21 -- This will fail due to UNIQUE constraint
22 ROLLBACK;
23
24 SELECT * FROM StudentEnrollments;
25 -- User A: Start transaction and lock the row
26 START TRANSACTION;
27 SELECT * FROM StudentEnrollments
28 WHERE student_name='Ashish' AND course_id='CSE101'
29 FOR UPDATE;
30
31 -- At this point, the row is locked until User A COMMITs or ROLLBACKs
32 -- User A can perform verification or updates safely
33 -- e.g., UPDATE StudentEnrollments SET enrollment_date='2024-07-03' WHERE student_name='Ashish' AND course_id='CSE101';
34
35 -- User B: In a separate session, attempt to update the same row
36

```



enrollment_id	student_name	course_id	enrollment_date
1	Ashish	CSE101	2024-07-01T00:00:00.000Z
2	Smaran	CSE102	2024-07-01T00:00:00.000Z
3	Vaibhav	CSE101	2024-07-01T00:00:00.000Z



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Run >

```
27
28 -- PART B
29 START TRANSACTION;
30 SELECT * FROM StudentEnrollments
31 WHERE student_name='Ashish' AND course_id='CSE101'
32 FOR UPDATE;
33
34 COMMIT;
35
36 -- Reset enrollment date for demonstration
37 UPDATE StudentEnrollments
38 SET enrollment_date='2024-07-01'
39 WHERE student_name='Ashish' AND course_id='CSE101';
40
41
42 -- User A: Start transaction and lock the row
43 -- PART C
44 START TRANSACTION;
```



enrollment_id	student_name	course_id	enrollment_date
1	Ashish	CSE101	2024-07-01T00:00:00.000Z



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Run >

```
50 UPDATE StudentEnrollments
51 SET enrollment_date='2024-07-03'
52 WHERE student_name='Ashish' AND course_id='CSE101';
53
54 -- User B: In separate session, attempts to update the same row
55 -- This will be blocked until User A commits
56
57 -- User A commits
58 COMMIT;
59
60 -- Now User B can proceed safely
61 -- After User B commits, data remains consistent; no race conditions
62 SELECT * FROM StudentEnrollments;
63
```

```
mysql> UPDATE StudentEnrollments
SET enrollment_date='2024-07-03'
WHERE student_name='Ashish' AND course_id='CSE101';
ERROR 1205 (HY000): Lock wait timeout exceeded; try restarting transaction
```

enrollment_id	student_name	course_id	enrollment_date
1	Ashish	CSE101	2024-07-03T00:00:00.000Z
2	Smaran	CSE102	2024-07-01T00:00:00.000Z
3	Vaibhav	CSE101	2024-07-01T00:00:00.000Z